MEADOWLANDS ESTATE WATER & HYDROLOGICAL ENGINEERING MANUAL

A Comprehensive Technical Guide to Mass-Conservation Fluid Dynamics, Reservoir Hierarchies, Automated 3-Stage Load Shedding, and Flood Mitigation

System: Hydrology Core v2.5.4 • **Target:** Estate Agronomy, Bio-Lab & Staff Support • **Published:** October 2026

1. Executive Summary & Core Physical Laws

The Meadowlands Hydrological Architecture is a fully closed-loop, physical fluid-dynamics simulation. Unlike generic idle farming mechanics where water is a simple consumable number, Meadowlands governs water through **Rigorous Mass Balance**: every liter of water originates from precipitation, deep subterranean artesian strata, or manual dew harvesting, and is tracked through storage buffers, gravitational conduits, soil saturation profiles, atmospheric evaporation, and percolation feedback.

THE FUNDAMENTAL ESTATE WATER EQUATION:
ΔStorage (Cistern + Tower + Aquifer) = [Precipitation + Subsoil Recharge + Dew Catchment] - [Furrow Discharge + Bio-Lab Cooling + Crew Hydration + Evapotranspiration + Seepage Loss]
- *Percolation Feedback: 30% of all furrow discharge and excess rainwater percolates back down into the Subterranean Aquifer.*
- *Moisture Scale: 1% soil moisture across 1 arable plot is equivalent to precisely 2.5 Liters of water volume.*

2. The Three Primary Reservoirs & Hydraulic Capacities

Water in Meadowlands is segmented across three specialized reservoirs, each operating at distinct hydrostatic pressures and geological elevations:

Reservoir	Role & Natural Source	Base Capacity	Max Tier (T5)	Dynamics & Key Operations
■■ **Rain Catchment Cistern**	Surface collection basin capturing atmospheric precipitation directly from rainfall & downpours.	1,500 L	75,000 L *(Mega-Basin)*	Inflow rate depends on weather condition (up to +120 L/min during storms). Transfers 100L batches to Gravity Tower via manual or worker lever.
■ **Elevated Gravity Tower**	Pressurized head-pressure tank on timber or steel framework supplying all furrow gates, labs, and crews.	800 L	50,000 L *(Mega-Citadel)*	Drives all estate water consumers. Governed by the Automated 3-Stage Load Shedding Ladder. Receives +15 L/s overflow surge when either Cistern or Aquifer reaches 100%.
■ **Deep Subsoil Aquifer**	Natural subterranean geological groundwater table situated beneath the estate subsoil.	3,000 L	80,000 L *(Abyssal Titan)*	Continuous natural artesian recharge (+8.0 L/min baseline). Receives +30% irrigation percolation. Supplies emergency float siphon valve during acute tower droughts.

3. Canal Network Progression & Engineering Physics

Water discharged from the Elevated Gravity Tower flows through the estate's canal network across unlocked arable plots (01–12). The physical efficiency of the canal depends on the installed engineering tier:

Tier & Name	Min Lvl	Coin Cost	Seepage Loss	Evap / Min	Saturation Cap & Protection

Tier 1: Hand-Dug Furrows	Lv 1	■ 0	**35% Seepage**	3.0%	100% (High risk of crop waterlogging & Root Rot if gates left open)
Tier 2: Clay-Lined Trenches	Lv 3	■ 18,500	**0% (Sealed)**	3.0%	100% (Impermeable compacted clay bed eliminates all subterranean seepage loss)
Tier 3: Timber Flumes	Lv 5	■ 65,000	**0% (Sealed)**	1.5%	**80% Ceiling (Root Rot Immune!)** - Elevated sluices baffle excessive flow
Tier 4: Sub-Surface Conduit	Lv 7	■ 180,000	**0% (Sealed)**	1.0%	**75% Ceiling** - Pressurized micro-drip delivery directly to plant roots

4. Soil Moisture Zones, Phenology & Automatic Shutoff Limiter

Each arable plot features real-time soil moisture percentage (0% to 100%). Crop phenology and final harvest yield are directly modulated by the soil moisture zone:

Zone	Moisture Range	Crop Growth Velocity	Crop Quality & Yield Effect	Recommended Operational Action
■ **Desiccated**	0% – 15%	0.25x (Frozen growth)	-50% harvest yield; high withering penalty	Immediate emergency irrigation required! Open master sluice gates.
■■ **Parched**	16% – 40%	0.70x (Sluggish growth)	Normal quality; vulnerable to midday heatwaves	Open sluice gates or pump cistern water to reach optimal zone.
■ **Optimal Range**	40.1% – 80%	**1.25x – 1.40x (Peak Speed)**	**+25% Crop Quality Bonus; +20% Bonus Yield**	Ideal agronomic balance. Calibrate Automatic Shutoff Limiter to ~75%.
■ **Saturated**	80.1% – 95%	1.00x (Standard)	Soil heavy; water film dampens respiration	Lock sluice gates to allow natural morning thermodynamic evaporation.
■■ **Waterlogged (Root Rot)**	95.1% – 100%	0.10x (Severe decay)	**Crop destruction within 3 minutes if unaddressed!**	Open Emergency Spillway gate immediately and clear silt grates.

The Automatic Saturation Limiter (Slider 50% – 95%):
On the Water page, estate managers can configure the automatic shutoff threshold (default: 75%). During live gate irrigation, once the average moisture across all unlocked plots reaches this calibrated ceiling, the master sluice gate snaps shut automatically, preventing accidental soil flooding and crop root rot while offline.

5. The Automated 3-Stage Load-Shedding Ladder & Emergency Siphon

The Elevated Gravity Tower acts as the pressurized heart of the estate. To protect critical high-value biological assets (such as incubated specimens in the Amber Bio-Lab) during severe droughts or player offline absence, the estate incorporates an automated 3-stage hydraulic load-shedding ladder:

Stage & Trigger Level	Status & Operational State	Crop Furrow Drain	Amber Bio-Lab Cooling	Worker Bunkhouse Station
Stage 0: Nominal State Tower > 200 Liters (>25%)	**Full Estate Operation** All municipal and agronomic lines active.	12.0 L/s (Full flow to open canal lines)	0.05 L/s per active chamber	0.015 L/s per active worker
Stage 1: Crop Conservation 80L – 200 Liters (10%–25%)	**Agronomic Load Shedding** Furrow gates forcibly isolated to preserve head.	**FORCIBLY SHUT (0 L/s)** Prevents tower drain to dirt.	0.05 L/s per active chamber (Preserved)	0.015 L/s per active worker (Preserved)

Stage 2: Crew Rationing 1L – 79 Liters (1%–10%)	**Staff Rationing Active** Domestic consumption halved.	0 L/s (Isolated)	0.05 L/s per active chamber (Preserved)	**Halved to 0.0075 L/s** Emergency canteen rations
Stage 3: Cryo-Standby 0 Liters (Total Depletion)	**Full Lockdown & Specimen Cryo-Lock** Specimens frozen in stasis; 0% decay.	0 L/s	**0 L/s (CRYO-STANDBY)** Bio-Lab specimens cannot die	0 L/s (Suspended)

■ ***THE AQUIFER SIPHON FEED VALVE (CONTINUOUS & EMERGENCY MODES):***
• ***Installation:*** *Crafted in the Workshop or installed directly via the Water Tab (Cost: 250 Coins).*
• ***Continuous Artesian Feed (Default):*** *Siphons clean groundwater from the Deep Aquifer at* ***+2.0 L/s*** *continuously whenever the Elevated Gravity Tower has headroom, maintaining 100% full capacity and constant pressurized head.*
• ***Emergency Reserve Mode (Toggleable):*** *For water conservation, can be switched to trigger only when the tower drops ≤ 80L (refilling up to 200L).*
• ***Total Drought & Offline Immunity:*** *As long as the Aquifer contains water, the Gravity Tower is continuously replenished and specimens never enter Cryo-Standby!*

6. Micro-Climate Profiles, Flash Floods & Emergency Spillway

Meadowlands features 8 dynamic meteorological conditions. Each weather pattern impacts precipitation inflow into the Cistern and thermodynamic evaporation from the arable furrows:

Weather Condition	Precipitation Inflow	Ambient Temp & Humidity	Evaporative Impact	Special Atmospheric Effect
■■ **Sunny / Clear**	0 L/min	24°C • 45% RH	Nominal (+2.2%/min)	Standard baseline field conditions.
■■ **Light Rain**	+45 L/min	18°C • 80% RH	Low (+0.6%/min)	Refills Rain Cistern continuously.
■■ **Downpour / Storm**	+120 L/min	15°C • 95% RH	Negligible (+0.1%/min)	Rapid cistern refill; soil saturates quickly.
■ **Rainbow Event**	+25 L/min	22°C • 60% RH	**0.0% / min (FROZEN)**	**Zero Evaporation across entire estate!** Perfect for crop hydration.
■ **Flash Flood Breach**	**+120 L/sec (Torrential)**	14°C • 98% RH	0% (Flooded)	**180s Hazard Timer!** Water surges over canal banks unless Spillway opened.

■■ ***EMERGENCY FLOOD SPILLWAY PROTOCOL:***
When a sudden river breach triggers a ***Flash Flood Hazard (180s countdown)****, torrential overflow threatens to submerge crops past 95% saturation, inflicting catastrophic Root Rot. Clicking* ***'Open Emergency Spillway'*** *engages the river bypass diversion gate, channeling 120 L/sec off-map and purging excess field water at -10%/sec until field equilibrium is restored.*

7. Autonomous Worker Shifts & Estate Operations

Hired farmhands automate critical water tasks during their scheduled day/night shifts, eliminating the need for constant micromanagement:

Shift & Worker Duty	Trigger Time / Condition	Automated Action Executed	Strategic Benefit
■ **Morning Field Hand**	Daily at 06:00 AM or whenever Tower < 200L	Opens master sluice gates for early morning irrigation; pumps aquifer if tower drops below nominal.	Crops enter peak morning photosaturation before midday evaporation intensifies.
■ **Night Watch Guard**	Night shift (22:00 – 05:00) or Cistern < 50L	De-silts canal grates, sweeps sediment, and harvests +20L condensed night dew into the cistern.	Prevents debris blockages; secures baseline water reserves during extended droughts.

■■ **Deploy Dew Screens**	Manual player action (Available on Water Tab)	Unfurls high-surface-area condensed moisture netting, capturing an immediate **+20 Liters** of fresh water.	Crucial early-game lifeline when starting out with Tier 1 furrows and 0 rain.

8. Master Agronomist Operational Checklist

STEP 1: INSTALL EMERGENCY SIPHON VALVE IMMEDIATELY
Visit the Water tab and install the Emergency Siphon Float Valve for 250 coins. This guarantees your Gravity Tower will never run dry while the Aquifer holds reserves.

STEP 2: CALIBRATE AUTO-SHUTOFF LIMITER TO 75%
Set the soil saturation limiter slider to 75%. This allows your crops to enjoy the 1.40x growth speed bonus while eliminating all risk of waterlogging rot.

STEP 3: UPGRADE CANALS TO TIER 2 BEFORE EXPANDING PLOTS
Tier 1 Earthen Furrows lose 35% of all released water into uncultivated turf. Upgrading to Clay-Lined Trenches (Tier 2, ■18,500) permanently stops all seepage loss.

STEP 4: TRIGGER OVERFLOW SURGE WITH FULL CISTERNS
*During heavy rainstorms, let your Rain Cistern reach 100% capacity. Once full, it triggers an instantaneous **+15 L/s Overflow Surge**, filling the Elevated Gravity Tower for free without spending coins on aquifer pumping!*